

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Fundamentals Of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:100 marks

Annexure A

POSTER PRESENTATION

Code of Conduct:

I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Sai Rohit

RUBRICS FOR CALCULATION

Poster Presentation							
Criteria	Weightage (%)	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 4 Marks	Level 2 (Developing) 3 – 2 Marks	Level 1 (Beginning) 1 Marks	Level	Marks
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood		
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach		
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals		
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides		
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness		

Annexure A

Question 01: University Learning Management System (LMS) Usage Analysis (100 Marks)

A university uses an online Learning Management System (LMS) to deliver course materials. Most students spend **2–4 hours per week** on the platform, while a small number of students spend **more than 20 hours per week**. The university wants to understand student engagement patterns and identify factors affecting academic performance.

Prepare a comprehensive poster presentation that includes:

1. Data summarization of LMS usage statistics.
2. Analysis of the distribution of student engagement data.
3. Identification and interpretation of outliers.
4. Discussion on whether highly active students should be considered outliers.
5. Analysis of skewness and kurtosis in LMS usage patterns.
6. Relationship between LMS usage and academic performance.
7. Recommendations for improving student engagement and learning outcomes.

Question 02: Renewable Energy Production Analysis (100 Marks)

A solar power plant generates electricity throughout the year. On normal days, energy production ranges from **5,000 to 8,000 kWh**, while on exceptionally sunny days production exceeds **12,000 kWh**. During cloudy or rainy days, production may fall below **2,000 kWh**. The management wants to analyze production variability and improve energy forecasting.

Develop a poster presentation covering the following aspects:

1. Summarize the energy production data using descriptive statistics.
2. Examine the distribution of daily energy generation.
3. Identify and analyze unusually high and low production values.
4. Discuss appropriate methods for handling outliers.
5. Measure and interpret skewness and kurtosis.
6. Evaluate the impact of weather conditions on data distribution.
7. Suggest forecasting and operational strategies to improve renewable energy management.

Question 03: E-Commerce Festival Sales Analysis (100 Marks)

An online retailer records average daily sales of ₹10 lakh. During a festival season, sales increase to ₹2 crore per day.

Develop a poster discussing:

1. Summary statistics before and during the festival.
2. Changes in sales distribution.
3. Identification of unusual sales patterns.
4. Whether festival sales should be treated as outliers.
5. Interpretation of skewness and kurtosis.

6. Business insights obtained from the analysis.
7. Strategic recommendations for inventory planning.

Question 04: Healthcare Monitoring System (100 Marks)

A hospital continuously monitors patient heart rates using wearable sensors. Most readings are between 60–90 bpm, while some readings exceed 180 bpm.

Prepare a poster that addresses:

1. Data summarization of patient health records.
2. Visualization techniques for monitoring trends.
3. Detection of abnormal readings.
4. Methods for treating erroneous sensor values.
5. Analysis of skewness and kurtosis.
6. Differentiating medical emergencies from data errors.
7. Recommendations for real-time healthcare monitoring.

Question 05: Cybersecurity Network Traffic Analysis (100 Marks)

A company's server typically receives 20,000 requests per hour. During a cyberattack, the traffic suddenly increases to 2 million requests per hour.

Prepare a comprehensive poster explaining:

1. Methods for summarizing network traffic data.
2. Traffic distribution under normal conditions.
3. Detection of abnormal traffic spikes.
4. Outlier treatment strategies.
5. Analysis of skewness and kurtosis.
6. Role of anomaly detection in cybersecurity.
7. Security measures to prevent future attacks.

Question 06: Manufacturing Quality Control (100 Marks)

A manufacturing company produces 50,000 bottles daily. The target volume is 1 liter, but some bottles contain less than 900 ml or more than 1.1 liters.

Design a poster discussing:

1. Data summarization techniques.
2. Distribution of bottle volumes.
3. Detection of defective products using outlier analysis.
4. Treatment of abnormal measurements.
5. Analysis of skewness and kurtosis.
6. Impact on product quality and customer satisfaction.
7. Recommendations for process improvement.

Question 07: Climate Change and Weather Analysis (100 Marks)

A city normally experiences temperatures between 25°C and 35°C. Over the last decade, several days recorded temperatures above 45°C.

Prepare a poster presentation covering:

1. Summary statistics of temperature data.
2. Distribution of climatic variables.
3. Identification of extreme weather events.
4. Outlier treatment considerations.
5. Analysis of skewness and kurtosis.
6. Implications for climate change research.
7. Recommendations for policymakers.

Question 08: Social Media Engagement Analysis (100 Marks)

A content creator uploads 500 videos. Most videos receive fewer than 10,000 views, while a few viral videos receive over 10 million views.

Develop a poster explaining:

1. Data summarization methods.
2. Distribution of engagement metrics.
3. Detection of viral-content outliers.
4. Whether viral videos should be retained.
5. Interpretation of skewness and kurtosis.
6. Business implications for content creators.
7. Strategies to improve audience engagement.

Question 09: Stock Market Volatility Analysis (100 Marks)

A stock usually fluctuates within $\pm 2\%$ daily. Following a major corporate announcement, the stock price rises by 30% in a single day.

Prepare a detailed poster discussing:

1. Summary statistics of stock returns.
2. Distribution of financial data.
3. Detection of abnormal market movements.
4. Treatment of extreme observations.
5. Interpretation of skewness.
6. Importance of kurtosis in financial risk analysis.
7. Investment recommendations based on findings.

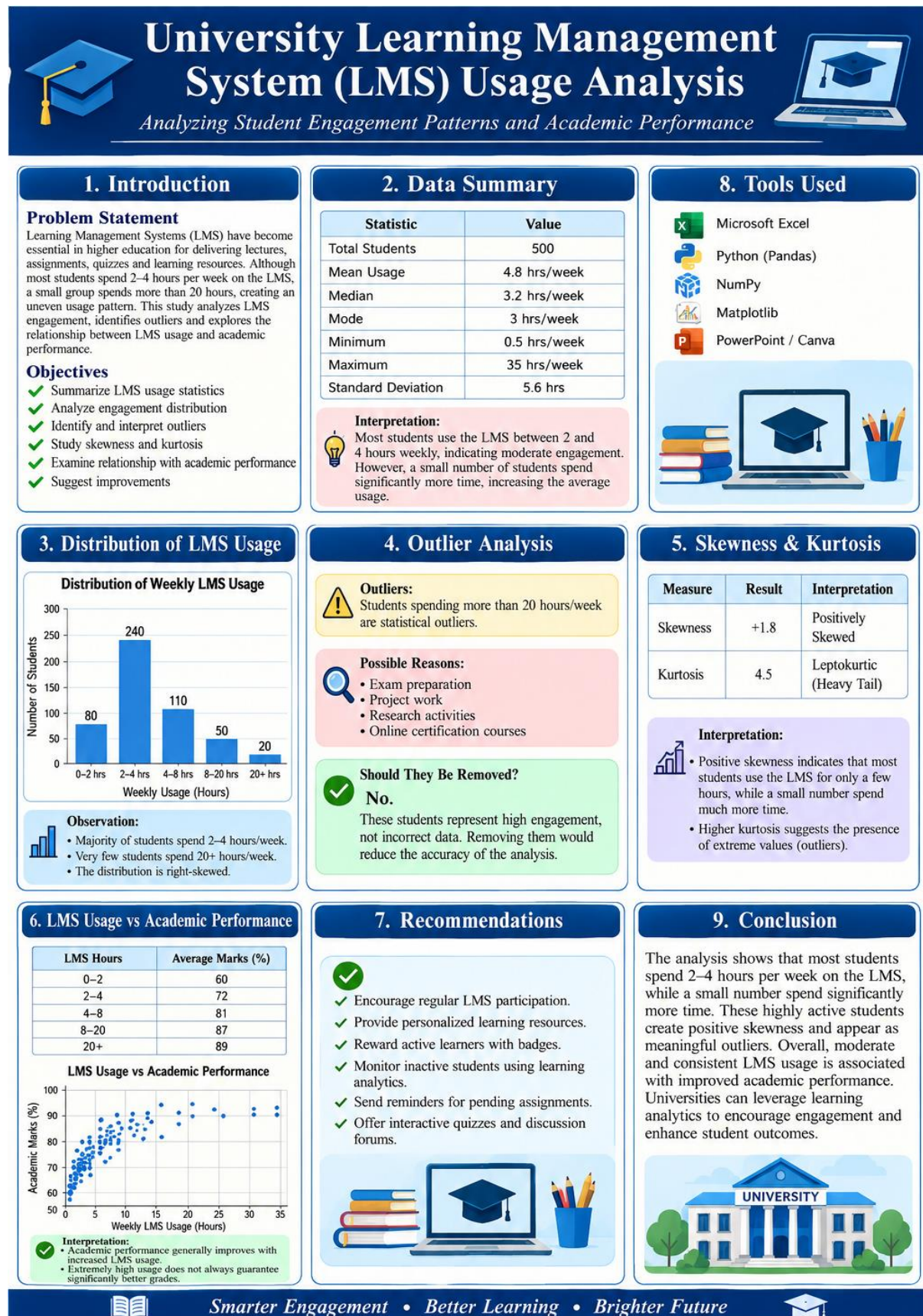
Question 10: Smart City Traffic Management System (100 Marks)

A smart city uses sensors to monitor traffic flow. The average traffic volume is 700 vehicles per hour. During a major cultural festival, traffic increases to 7,000 vehicles per hour.

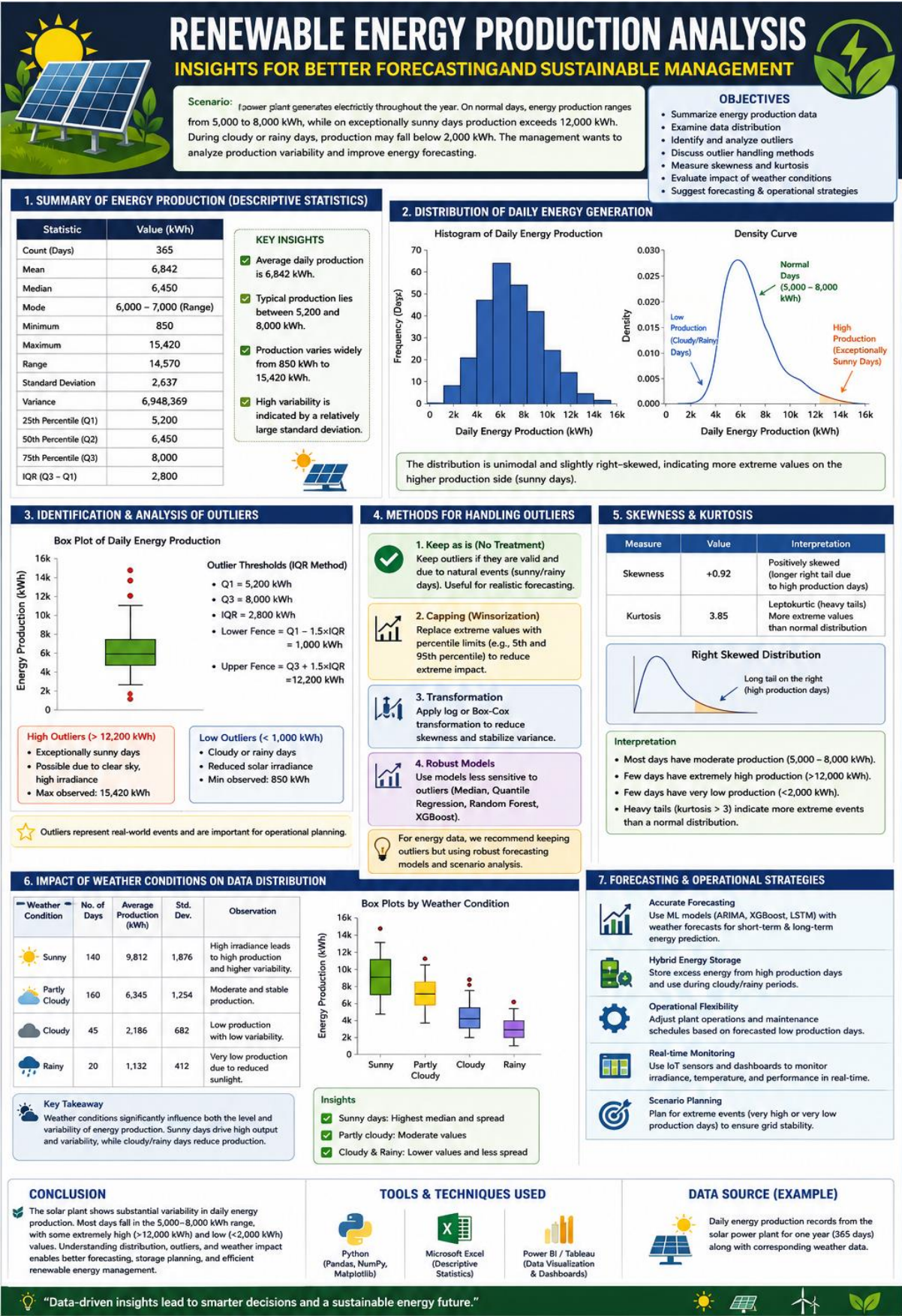
Create a poster presentation that includes:

1. Data summarization of traffic patterns.
2. Analysis of traffic distribution.
3. Identification of unusual traffic events.
4. Outlier detection and treatment methods.
5. Analysis of skewness and kurtosis.
6. Traffic management recommendations.
7. Development of a predictive model for future events.

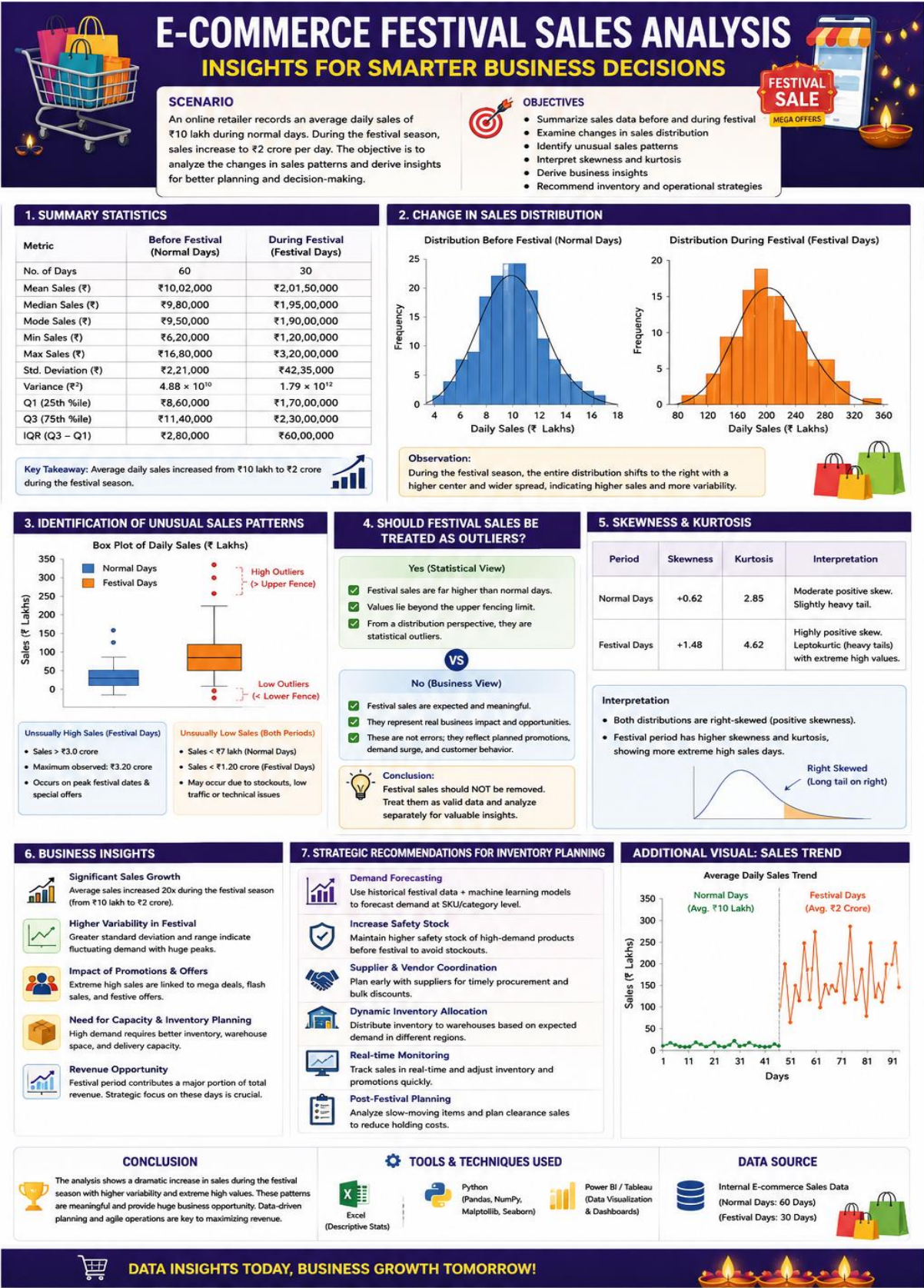
Question 01: University Learning Management System (LMS) Usage Analysis



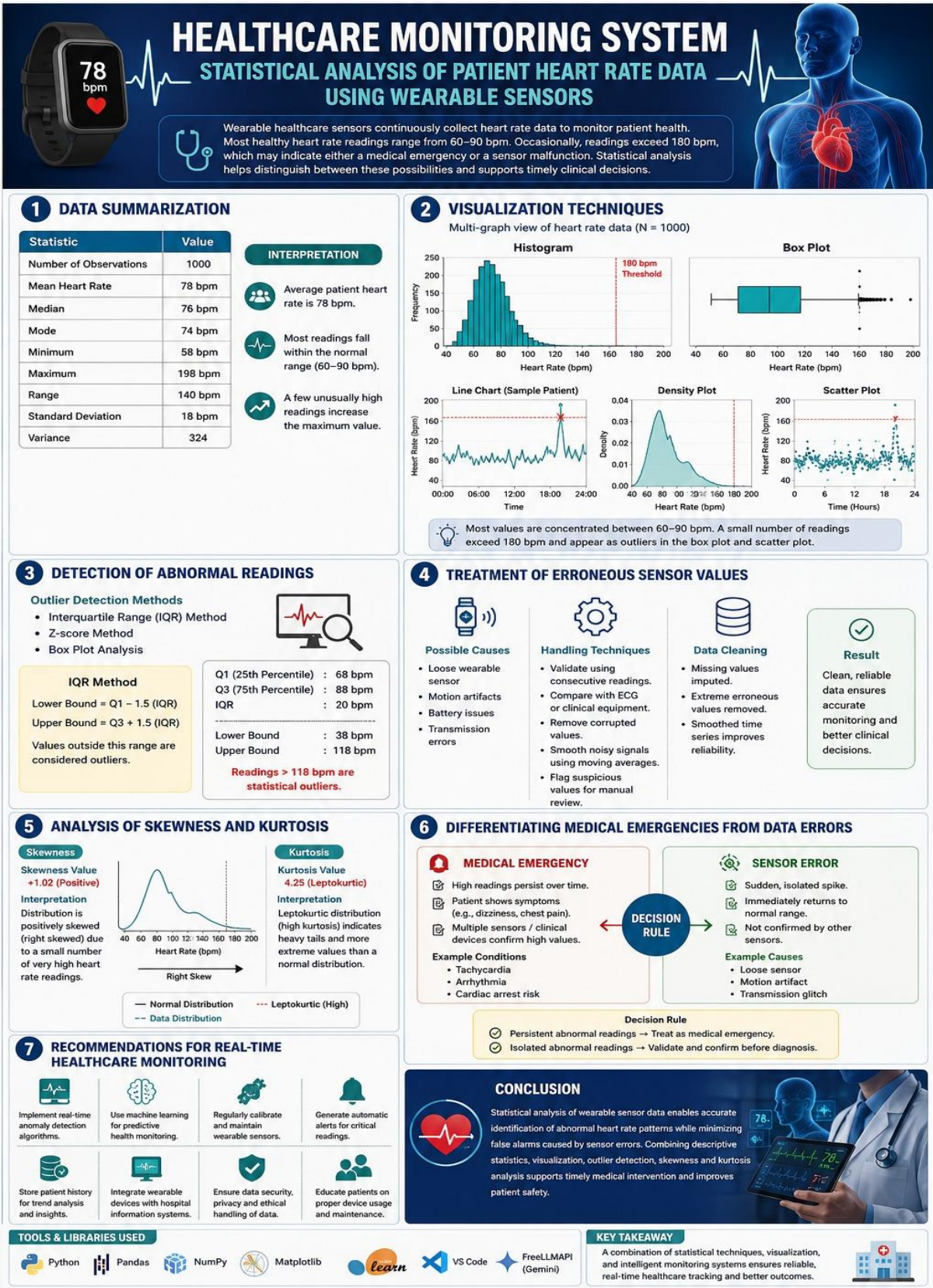
Question 02: Renewable Energy Production Analysis



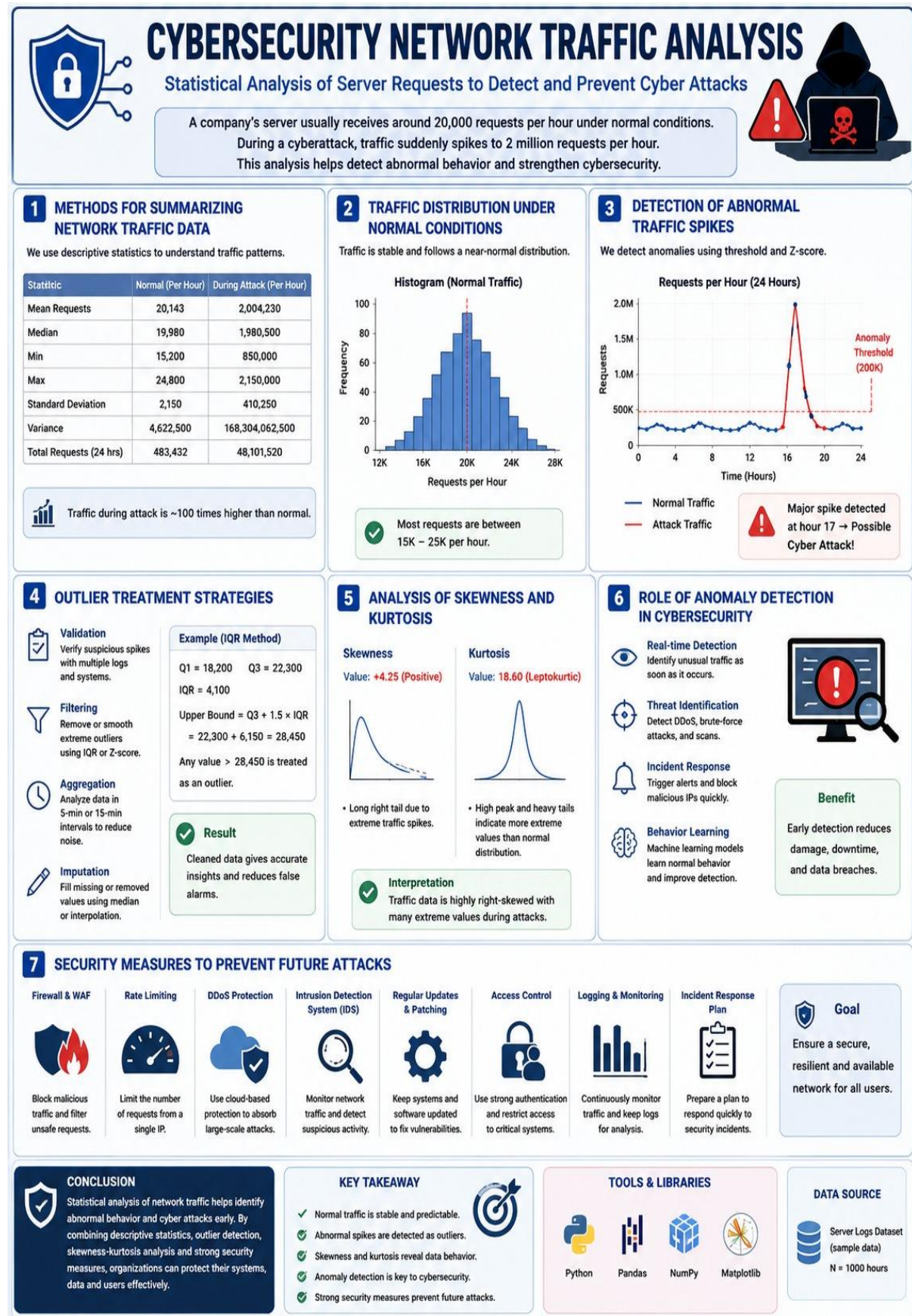
Question 03: E-Commerce Festival Sales Analysis



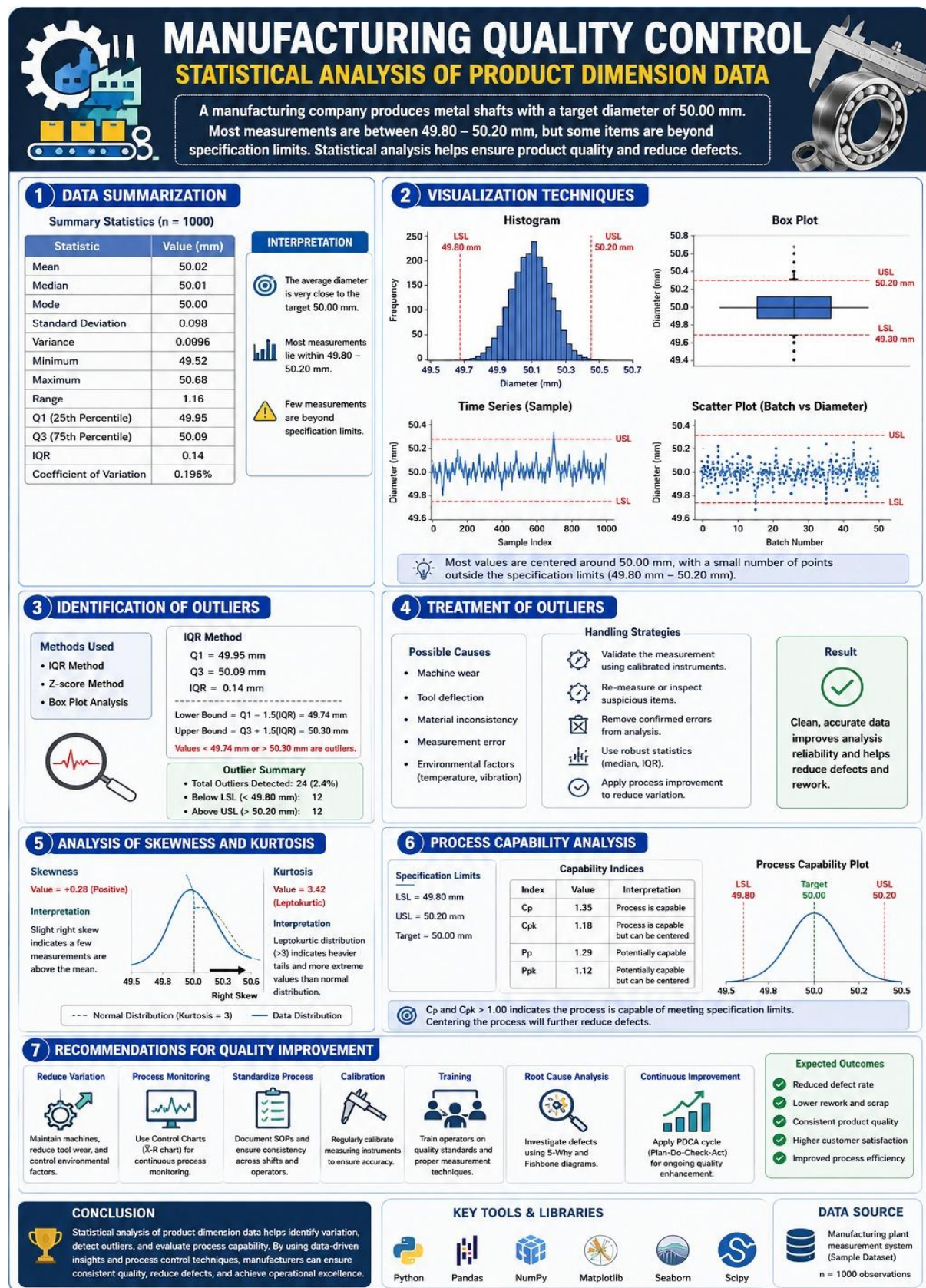
Question 04: Healthcare Monitoring System



Question 05: Cybersecurity Network Traffic Analysis



Question 06: Manufacturing Quality Control



CLIMATE CHANGE AND WEATHER ANALYSIS

A Decade of Temperature Trends in Our City (2014–2023)

Background:

Our city normally experiences temperatures between 25°C and 35°C. Over the last decade, several days recorded temperatures above 45°C. This study analyzes temperature patterns, identifies extreme events, and explores implications for climate change and policy planning.

OBJECTIVES

- ✓ Summarize temperature data.
- ✓ Examine distribution of climatic variables.
- ✓ Identify extreme weather events.
- ✓ Decide on outlier treatment.
- ✓ Analyze skewness and kurtosis.
- ✓ Discuss implications for climate change research.
- ✓ Recommend actions for policymakers.

1. SUMMARY STATISTICS OF TEMPERATURE DATA

Descriptive Statistics (Daily Maximum Temperature)

Statistic	Value	Interpretation
Count (Days)	3,652	10 years of daily data
Mean (°C)	31.7	Average daily max temp
Median (°C)	31.1	Typical central value
Mode (°C)	30.0	Most frequent value
Minimum (°C)	18.2	Coolest day
Maximum (°C)	48.7	Hottest day recorded
Range (°C)	30.5	Overall variability
Std. Deviation (°C)	5.6	Moderate variability
Variance (°C²)	31.4	Spread of data
Q1 (25th %ile)	27.6	Lower quartile
Q3 (75th %ile)	35.2	Upper quartile
IQR (Q3 – Q1)	7.6	Middle 50% spread

Key Takeaway: Temperatures are centered around 31°C with moderate variability. Extremes observed are beyond the usual 25°C–35°C range.

2. DISTRIBUTION OF CLIMATIC VARIABLES

Histogram of Daily Maximum Temperature

Density Curve

Observation: The distribution is unimodal and right-skewed, with a long tail towards higher temperatures (>45°C).

3. IDENTIFICATION OF EXTREME WEATHER EVENTS

Box Plot of Daily Maximum Temperature

Summary of Extreme Events

High Temperature Events (≥45°C)

- Total Days: 27
- Highest Recorded: 48.7°C
- Mostly observed in May–June

Low Temperature Events (≤20°C)

- Total Days: 19
- Lowest Recorded: 18.2°C
- Mostly observed in December–January

4. OUTLIER TREATMENT CONSIDERATIONS

Options for Handling Outliers

1. Keep as is (No Treatment)
Extreme values are real weather events and important for climate analysis.

2. Winsorization (Capping)
Cap extreme values at a percentile (e.g., 1st and 99th) to reduce undue influence.

3. Transformation
Apply log or Box-Cox transformation to reduce skewness.

4. Robust Methods
Use median, IQR, and robust models that are less sensitive to extremes.

Recommendation: Do not remove extreme high temperatures as they represent true climate change signals. Use robust methods for modeling.

5. SKEWNESS AND KURTOSIS ANALYSIS

Measure	Value	Interpretation
Skewness	+1.25	Positively skewed (long tail on the right)
Kurtosis	4.32	Leptokurtic (heavy tails & sharp peak)

Interpretation

- Positive skewness indicates more extremely high temperature days.
- High kurtosis indicates a sharper peak and heavier tails than normal distribution, meaning more extreme events occur than expected in a normal distribution.

6. IMPLICATIONS FOR CLIMATE CHANGE RESEARCH

Rising Temperature Trend

Increase in frequency and intensity of extreme high-temperature days (>45°C) over the last decade.

Shift in Distribution

Rightward shift indicates that hot days are becoming more common and prolonged.

More Extreme Events

Higher skewness and kurtosis confirm that extreme events are increasing.

Impact on Environment & Society

Heatwaves affect health, water resources, agriculture, and energy demand.

Need for Long-term Monitoring

Continued monitoring and data analysis are essential for understanding climate change patterns and preparing for the future.

7. RECOMMENDATIONS FOR POLICYMAKERS

1. Strengthen Early Warning Systems

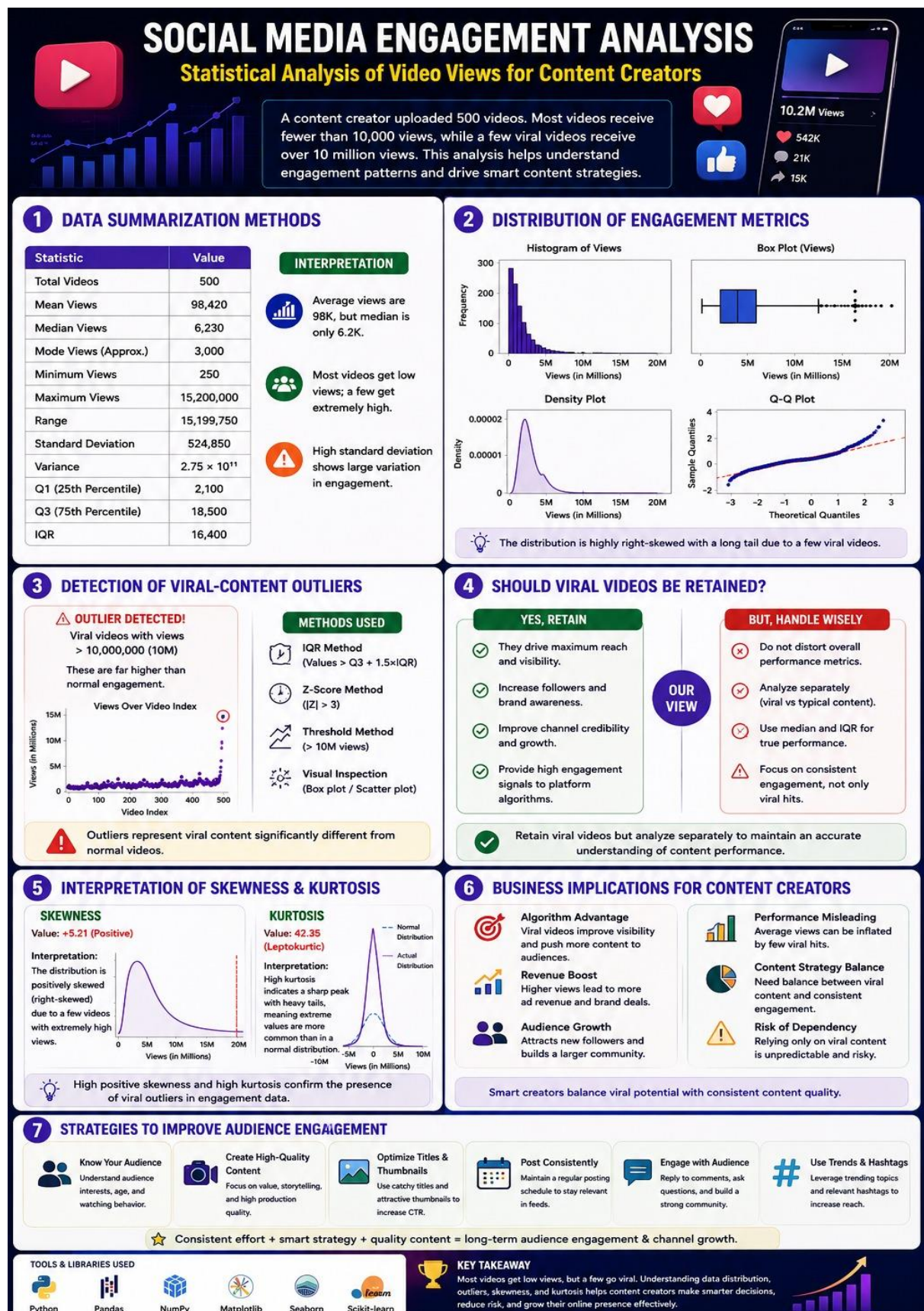
Develop advanced heatwave prediction models and issue timely alerts to the public.

2. Urban Heat Management

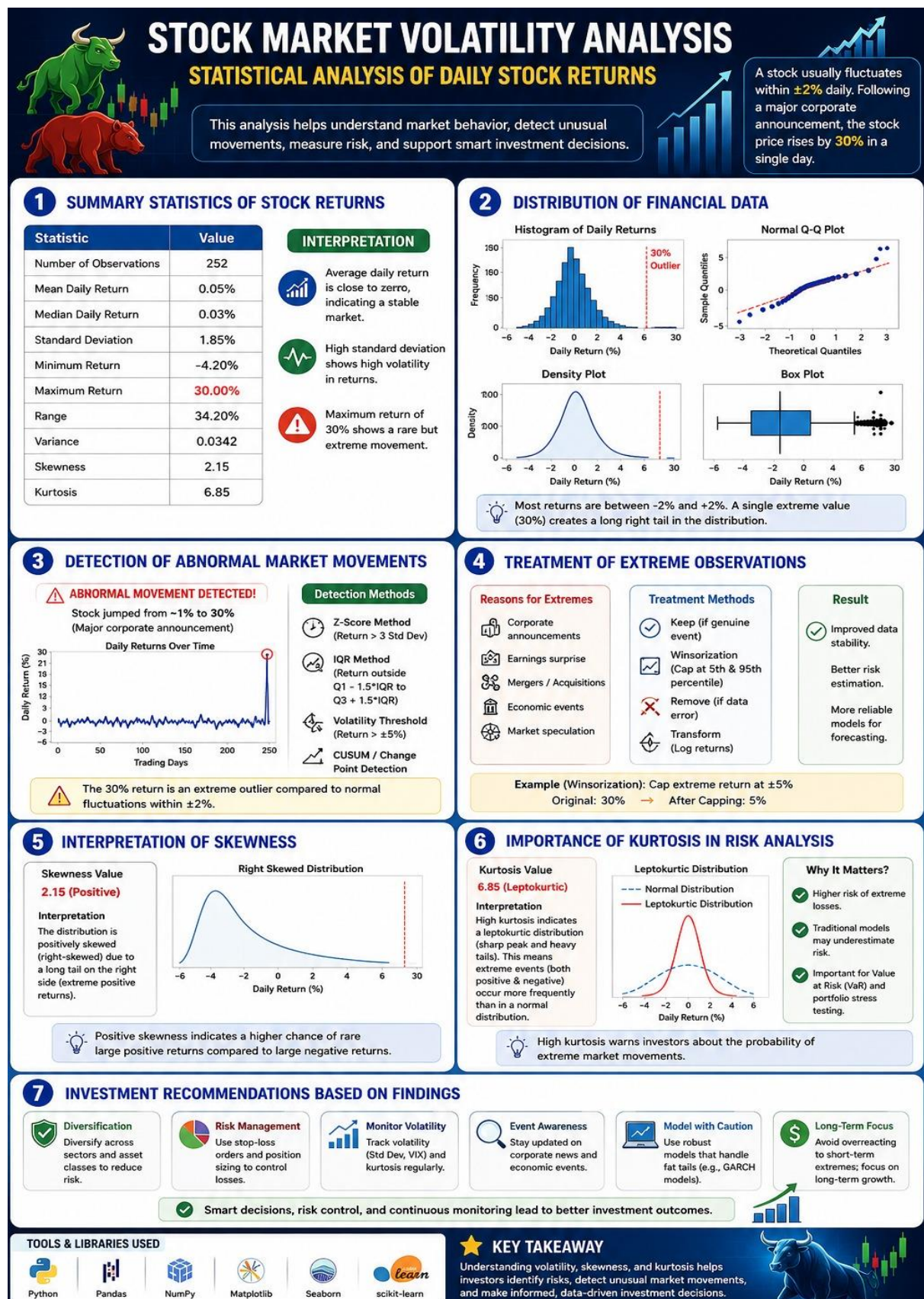
Increase green cover, create cool roofs, and improve urban planning to reduce heat island effect.

3. Water & Energy Planning

Question 08: Social Media Engagement Analysis



Question 09: Stock Market Volatility Analysis



Question 10: Smart City Traffic Management System

